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Assessment Paper AY 2022 Semester 2 E211 MUE Main (1)

Section A (11 Questions)

Q1. Which one of the below statements is **NOT** an assumption of M/D/2 queuing model?

- Number of channels is 2
- Number of service phases is 1
- Service time is random (Correct Option)
- Arrival pattern follows Poisson distribution

Q2. Which one of the following problems would likely **NOT** require the use of simulation?

- Studying visitor flow and queues at various rides in a theme park
- Determining the most efficient layout of a large distribution centre
- Predicting the impact on a complex production process when resource levels change
- Deciding whether to open more physical stores or expand online presence (Correct Option)

Q3. The Minimum Cost Flow problem is a generalization of all the following network problems **except**

- Shortest path problem
- Maximum flow problem
- Travelling salesman problem (Correct Option)
- Transportation problem

Q4. Applying Nearest Neighbour Method to the network in Table 1, the routing solution (**starting node = B**) would be

Table 1 Travel Time between Nodes

	A	B	C	D	E
A		19	21	30	45
B	19		35	22	25
C	21	35		33	40
D	30	22	33		28
E	45	25	40	28	

- B – D – C – E – A
- B – C – D – E – A – B
- A – B – D – C – E – A
- B – A – C – D – E – B (Correct Option)

Q5. Alex wants to forecast the demand for a potential new product. He consults the sales personnel, and they would explain to him how well the product may be received by the market. This form of forecasting is known as

- Grassroots forecasting (Correct Option)
- Delphi method
- Focus group
- Customer survey

Q6. *Show your workings clearly*.

A distribution network diagram is shown in Figure 2a. The nodes represent locations and numbers on the arcs represent the distances between locations.

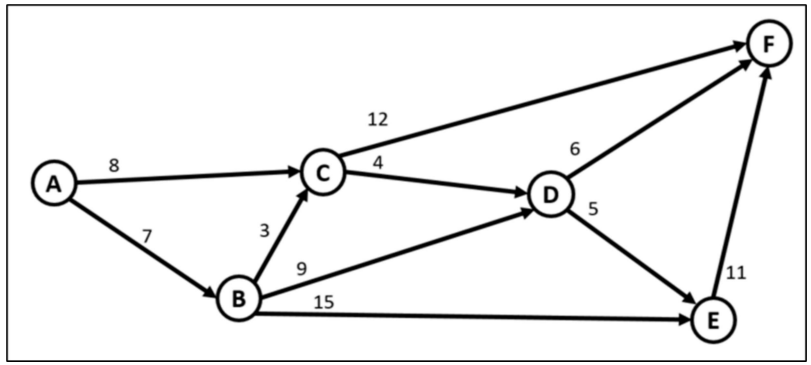

Figure 2a Network Diagram

Q6a. Table 2a below shows a **partial** method used to find the shortest path in Figure 2a from node A to each of the other nodes.

Table 2a Table to Solve Shortest Path Problem

Iteration	Solution Set	Arcs	Total Distance
Iteration 1	A	A → B A → C	7 8
Iteration 2	A B (7)	A → C B → C B → D B → E	8 **W** 16 22
Iteration 3	A B (7) C (8)	B → D B → E C → D C → F	15 22 12 20
Iteration 4	…	…	…
…	…	…	…

Q6a(i). Name the algorithm used in Table 2a to solve the shortest path problem.

Q6a(ii). Calculate **W** as shown in Table 2a.

Q6a(iii). Which is the next node to be added into the solution set at **Iteration 4**? Explain.

Q6a(iv). List down the shortest path from node A to node E and state the total distance.

Q6b. A Linear Programming (LP) model is formulated to solve the shortest path problem in Figure 2a. Assume that the decision variable Xij represents whether the arc from node i to j is on the shortest path. (i, j = A, B, C, D, E or F)

For example, XAC represents whether the arc from node A to node C is on the shortest path.

Q6b(i). What type of decision variable is used to formulate the above LP model?

Q6b(ii). Construct the net flow constraints for the following nodes based on the decision variables defined above.

a) Node D

b) Node F

Q6b(iii). If the shortest path must pass through node C, formulate this requirement as a constraint.

Q6c. In one city, a product has to be distributed to a number of stores. Table 2b shows the distance between each pair of nodes. Warehouse is at **node F**.

Table 2b Travel Distance Between Nodes

From	To (Travel distance in kilometers)				
	F	G	H	J	K
F	-	24	18	20	29
G	24	-	22	31	38
H	18	22	-	25	31
J	20	31	25	-	26
K	29	38	30	26	-

Clark and Wright savings method is used to plan a delivery route. Table 2c shows the **savings** in distance between 2 stores ranked in descending order.

Table 2c Savings Ranked in Descending Order

Rank	Savings	From	To
1	23	J	K
2	?	G	H
3	16	H	K
4	15	G	K
5	13	G	J
6	13	H	J

Q6c(i). Calculate the Savings between nodes G and H.

Q6c(ii). Based on the Savings table in Table 2c, at Rank 1, a sub-tour F-J-K-F should be formed. What step should be carried out next at Rank 2?

Q6c(iii). State the complete route based on Clark and Wrights Savings Method and calculate the total travel distance.

Q7. *Show your workings clearly and round off your answer to* ***3 decimal places*** ***where necessary.***

Kate recently joined a new company that develops high-tech household products. Her role in the company is demand planning.

Q7a. The company launched a new robot cleaner a few months ago. To predict customer demand, Kate collected past sales data and carried out a forecast using Weighted Moving Average (Table 3a).

Table 3a Sales Data from Past 6 Months

Month	Sales Figure
1	36
2	38
3	29
4	45
5	42
6	36

Q7a(i). Based on weights of 1, 3 and 5, calculate the forecast for **Month 7.**

Q7a(ii). If Kate wants the forecast to be less responsive to changes in the most recent data, what can she do?

Q7b. For another product, Kate collected the monthly sales figures for the past 10 months (Figure 3a). What forecasting method should Kate apply for this product? Give a reason.

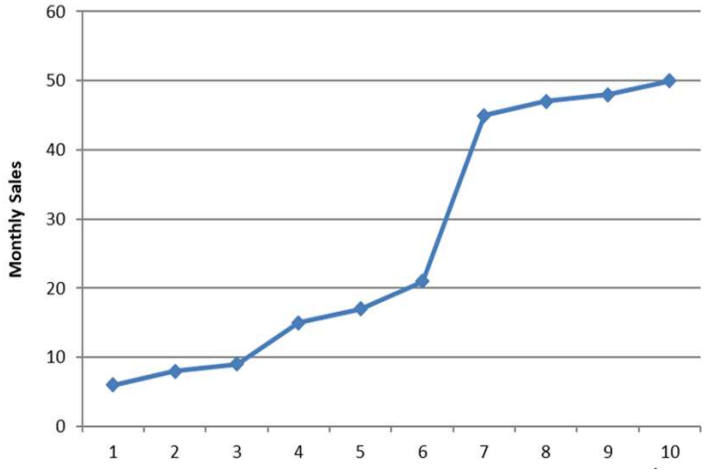

Figure 3a Monthly Sales Figure

Q7c. Kate developed an Excel template (Table 3b) to help her do the demand forecasting for a 3rd product.

Table 3b Partial Workings for Forecasting Demand

Input Data			Seasonal Index Computation					Forecast	
Month	Actual Value (Y)	Time Period (X)	Moving Average	Centred Moving Average	Seasonal Ratio	Seasonal Index	Deseasonalized value	Deseasonalized forecast	Seasonalized forecast
2018Q1	432	1				0.843	512.330	560.962	473.007
2018Q2	564	2	682			0.894	630.564	634.891	567.870
2018Q3	656	3	**(A)**		0.915	0.935	701.910	708.820	662.458
2018Q4	1076	4		786	**(C)**	1.328	810.385	782.749	1039.307
2019Q1	710	5	904	862.25	0.823	0.843	**(E)**	856.679	722.357
2019Q2		6	992.5	948.25	0.886	0.894	939.138	930.608	832.370
2019Q3	990	7	1068.5	**(B)**	0.961	0.935	1059.285	1004.537	938.832
2019Q4		8	1138	1103.25	1.296	1.328	1076.998	1078.466	1431.948
2020Q1	1014	9	1195.5	1166.75	0.869	**(D)**	1202.553	1152.395	971.707
2020Q2		10	1263	1229.25	0.909	0.894	1249.948	1226.324	1096.870
2020Q3	1220	11				0.935	1305.382	1300.253	1215.207
2020Q4	1700	12				1.328	1280.348	1374.182	1824.590
				Intercept for 'Deseasonalized Value'			**487.033**		
				Slope for 'Deseasonalized Value'			**73.929**		

Q7c(i). How many periods of data are used to calculate the values in the Moving Average column?

Q7c(ii). Calculate the Moving Average labelled as **(A)** in Table 3b.

Q7c(iii). Calculate the Centred Moving Average labelled as **(B)** in Table 3b.

Q7c(iv). Calculate the Seasonal Ratio labelled as **(C)** in Table 3b.

Q7c(v). Determine the Seasonal Index labelled as **(D)** in Table 3b.

Q7c(vi). Calculate the Deseasonalized Value labelled as **(E)** in Table 3b.

Q7c(vii). Calculate the Seasonalized Forecast for **2021 Quarter 2**.

Q8. *Show your workings clearly and round off your answers to* ***three decimal places*** ***where necessary.*** *Use ^ symbol to represent the power notation.*

Jane runs a self-service print shop. Customers walk into her shop according to a Poisson process. On average, the time between customer arrivals is 12 minutes. According to data collected, the average service time is 10 minutes with a standard deviation of 6 minutes.

Q8a. Calculate the average customer arrival rate λ, in per hour.

Q8b. Assuming that there is 1 printer in the shop, state the appropriate queuing model (using Kendall's notation) for the queuing process.

Q8c. Calculate the average service rate, μ in per hour.

Q8d. Calculate the average number of customers in the queue, Lq.

Q8e. The printer interface has been upgraded and customers find it easier to use. The average service time remains unchanged but the variation in service times is greatly reduced. Explain how this would affect the performance measures of the queuing system.

Q8f. Jane's printing business has grown over the past year and she decides to add more printers. Suppose customer arrival rate has increased to 14 per hour and service rate is 6 per hour.

Q8f(i). What is the minimum number of printers required for the queuing process in her shop to be in a steady state?

Q8f(ii). Given the average number of customers in queue, Lq is 2.14, calculate the average number of customers in her shop, Ls.

Q8f(iii). Assume that June has **3 printers** in the shop. Using Table 4a where n is the number of customers in the system, calculate the probability of having **at least 2 customers in the queue.**

Table 4a Partial Workings of M/M/3 Queuing Model

n	$\frac{(\lambda/\mu)^n}{n!}$	$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	0.084	0.378
1	2.182	0.183	0.275
2	2.380	0.200	0.200
3	1.731	0.145	0.145
4	0.944	0.079	0.106
5	0.412	0.035	0.077
6	0.150	0.013	0.056

Q8g. Jane noticed that some customers would not come into the shop if the queue was long. What do you call this behaviour?

Q9. Jack is using FlexSim process flow simulation to study robot material handling in a facility. Figure 5a shows a snapshot of the partial model during simulation.

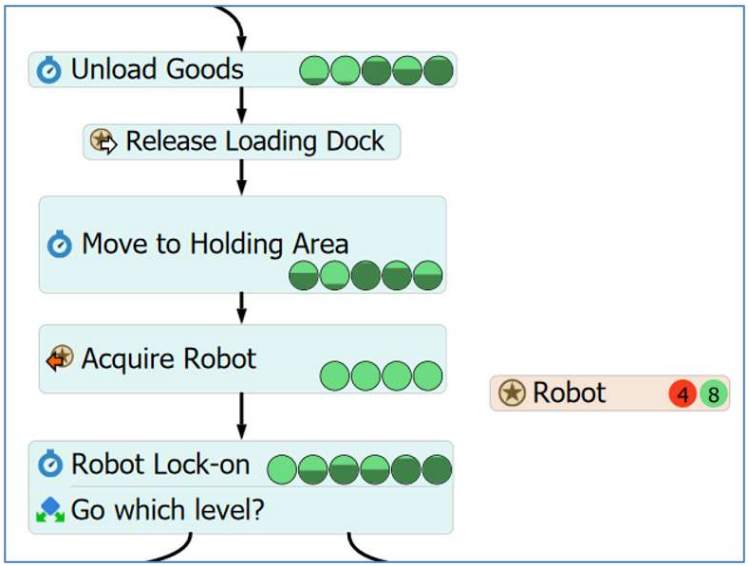

Figure 5a FlexSim Process Flow Simulation

Q9a. Referring to Figure 5a, state:

Number of Robots deployed = 8; eight

Number of items in the process of moving to holding area = 5; five

Q9b. Suppose Jack made a mistake and did not insert 'Acquire Robot' activity in the model. How would this affect the simulation?

Q9c. Name the activity from FlexSim Process Flow library used to model 'Go which level?'. What is its function?

Q9d. A new improved Robot lock-on mechanism has been designed. As the mechanism is new, Jack decides to model the lock-on timings with a Triangular distribution. What parameters are required for this distribution?

Q9e. Figure 5b shows the statistics page of the Robot resource. Answer the following questions:

Input

Current
3239.00

Output

Current
3231.00

Content

Current	Min	Max	Avg
8.00	0.00	8.00	6.77

Staytime

Min	Max	Avg
0.27	109.12	19.92

Request Input

Current
3243.00

Request Output

Current
3239.00

Request Content

Current	Min	Max	Avg
4.00	0.00	17.00	2.08

Request Staytime

Min	Max	Avg
0.00	44.09	6.11

Resource Available

Current	Min	Max	Avg
0.00	0.00	8.00	1.23

Figure 5b Robot Resource Statistics

Q9e(i). State the average waiting time for a Robot.

6.11

Q9e(ii). How much space is to be allocated at the waiting area for Robot? Assume the value should be equal to the maximum queue length.

17; 17 items

Q9e(iii). Calculate the Robot resource average utilization.

Q10. Jane works as a process leader in the packing department of a production facility. The packing process consists of two stations – 'Sealing_and_Labelling' and 'Inspection'. To study the packing process, she builds a 3-D simulation model in FlexSim. Figure 6a shows the simulation at the end of a run.

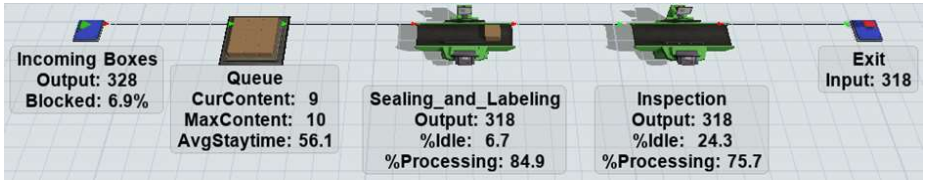

Figure 6a Simulation of Packing Process

Q10a. State the following performance measures:

Number of boxes inspected = 318

Utilization of Sealing_and_Labeling = 84.9; 84.9%

Q10b. The 'Max Content' of Queue object is currently 10. Explain how it would affect the results of the simulation if this value is increased to 20.

Q10c. Identify the bottleneck of the model in Figure 6a. Give a reason.

Q10d. State **two** ways the above simulation would change if the bottleneck identified is resolved.

Q10e. Jane was told that it may not be necessary to inspect all the boxes. After Sealing_and_Labeling, 70% of the boxes would be sent for Inspection and 30% would bypass Inspection and be sent to Exit directly. The modified FlexSim model is shown in Figure 6b. State what she needs to set in the 'Send to Port' option for Sealing_and_Labeling.

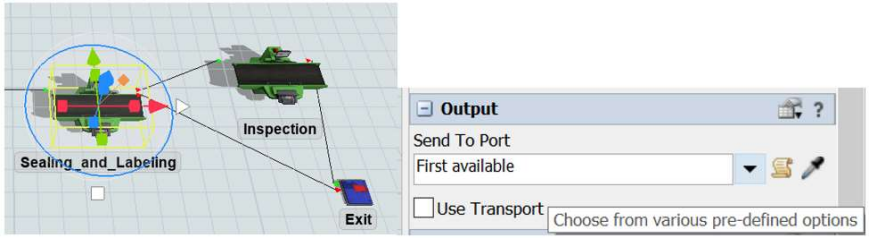

Figure 6b Modified Packing Simulation Model

Q10f. Jane would like to use the Experimenter in FlexSim to generate average waiting time results for 30 runs. Referring to Figure 6d, explain what Jane needs to do before she could run the Experimenter.

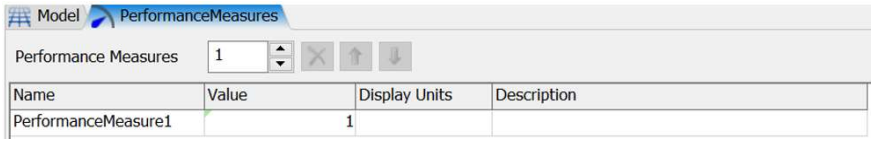

Figure 6d Performance Measures Table

Q10g. Suggest **two disadvantages** of using simulation for the study.

Q11. Peter constructed the Patient Flow in FlexSim Healthcare shown in Figure 7a to simulate a registration and consultation process in a clinic.

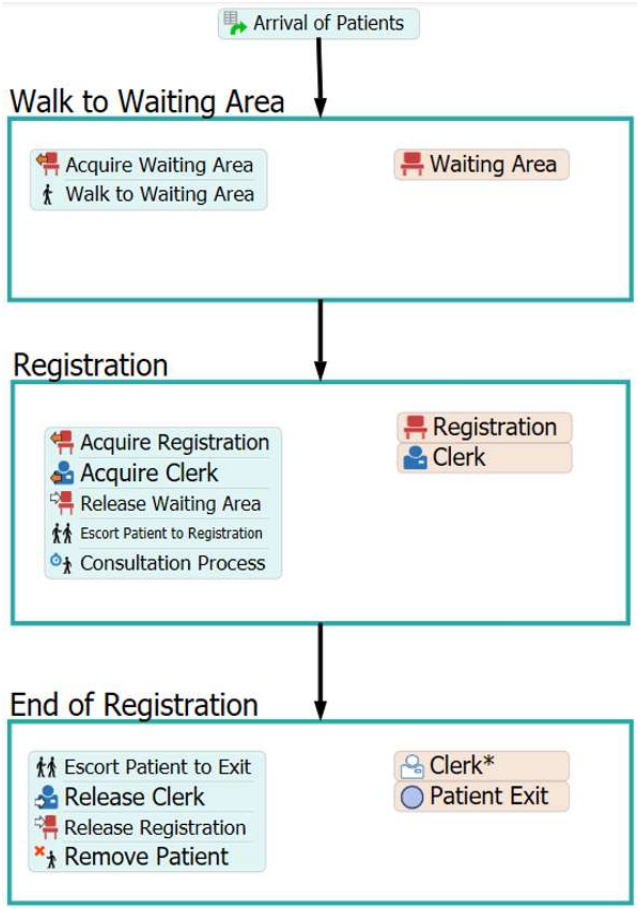

Figure 7a Patient Flow Model in FlexSim Healthcare

Q11a. Given the patient arrival schedule below (Figure 7b), where should Peter enter this data in the FlexSim Healthcare model?

Start (DD:)HH:MM:SS	End (DD:)HH:MM:SS	Quantity	Name
08:00:00	08:30:00	5	
08:30:00	09:30:00	3	
09:30:00	10:00:00	3	

Figure 7b Patient Arrival Schedule

Q11b. Identify two roles of the Clerk based on the Patient Flow Model.

Q11c. **'Registration'** is a type of Location Resource. Briefly describe how to link the Registration object in the 3D model to the Registration resource in Process Flow.

Q11d. Peter plans to create a Milestone-Milestone Times chart at a Dashboard to measure the average time that patients need to wait at the waiting area before being served. Where should Peter insert the Milestones within the Patient Flow model created in Figure 7a? (Milestone can be found in the Library of Healthcare Process Flow as shown in Figure 7c)

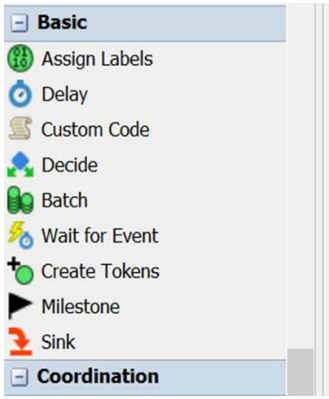

Figure 7c FlexSim Healthcare Process Flow Basic Tools

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